PROBLEM SET / 02

Reproducible agentic analytics for civic records

Verified million-record backend, failed local-agent evaluation

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Developed after the original submission and grading dates. Status: In progress.

Abstract

Natural-language civic questions hide choices about time boundaries, complaint families, geography and missing values. This implementation places those choices in a strict analytical contract. An agent handles conversation; a Python service validates specifications, computes results and derives artifacts through JSON CLI and seven MCP tools. [1] [2]

Version 0.7 acquired 2,133,268 unique NYC 311 requests. Nine pre-study executions reconciled that count through normalization, immutable Elasticsearch 9.5.5 indexes and independent SQLite references. Five analytical families and complete CSV selections agreed. The eighth finite workload measured query medians of 29-77 ms and a 9,728-row export in 1.060 s. These are backend observations, not conversational latency or maximum capacity. [3] [4] [5]

Contribution and result boundary

The ninth run verified all three Maps cases after eight retained failures. An independent raw-capture audit matched every cell in four 9,728-row exports. However, the frozen Qwen3-1.7B candidate passed only one of 120 executed and reviewed trials (0.83%); all nine chained analyses failed. Three additional study core preflights passed, isolating backend qualification from agent reliability. [4] [15] [16] [17]

Backend and Maps checks passed within the measured scope. Agent quality FAILED the fixed 90% and every-question requirements; the sole passing answer was an unsupported-request refusal. The conversational release is unqualified. Billion-record capacity remains a design target.

1. Research framing and semantics

Text-to-query research separates understanding a question from executing its program. Spider evaluates complex queries and generalization across database schemas. This project instead uses a fixed civic catalog and Elasticsearch; the reported study is not a Spider reproduction or a cross-domain accuracy claim. [9]

Elastic documents search_after with point-in-time contexts for stable deep pagination. The adapter follows that model for bounded record retrieval; actual correctness and performance evidence is supplied by the separate retained execution receipts. Documentation alone is not empirical validation. [4] [5] [10]

Formal analytical object

Let D_v be an immutable observed dataset and q=(v,F,I,G,M,Z) its validated specification: version, non-time filter, half-open interval, grouping, metric definitions and timezone. Let C(q) contain the records in D_v satisfying F and I. Preview, CSV and map derive from that saved cohort, rather than independently reconstructing the natural-language question. [2]

```
n(g,I) = cohort records in group g
rate(g,I) = n(g,I) / calendar_exposure_days(I,Z)
change(g) = rate(g,I_after) - rate(g,I_before)
mean_closure(g) = sum(valid durations) / count(valid durations)
```

Mean closure is administrative created-to-closed elapsed time, not first response or verified resolution. Its denominator can differ from both total and closed-request counts. Exposure days use the requested timezone, whereas closure uses elapsed hours. Reporting rates describe observed requests and reporting behavior, not causal agency performance. [2] [11]

Research question: can a bounded analytical contract support correct conversational answers and matching artifacts across agents at usable interaction cost? Real-corpus backend agreement answers only one part of that question.

2. Method and implemented architecture

A request selects records, aggregate or compare_periods and binds it to a dataset version. Bounded Boolean filters, category families, geographic predicates and explicit time ranges are supported. Unsupported combinations return typed errors. A saved result carries the normalized specification, exact index identity and execution flags. [2]

Phase	Service responsibility
Discover	Expose catalog, field meanings, observed scope, limits and request examples.
Validate	Resolve time boundaries; reject unsupported operations and unqualified comparisons.
Execute	Compile filters and aggregations; persist result, provenance and completeness.
Inspect	Return bounded previews with full matching totals and approximation labels.
Derive artifact	Preserve saved cohort and selected groups; record excluded geography and hashes.

Time, coverage and missingness

Intervals are start-inclusive and end-exclusive. The captured window covers April 1 to November 1, 2025 in America/New_York. Reconciled capture, normalization and frozen-index counts qualify comparison periods within that observed corpus; no captured records were dropped. The certificate explicitly retains population_complete=false and transactional_source_snapshot=false; a provider revision match is not proof of either stronger property. [3] [4]

Administrative closure is elapsed creation-to-closure time, not first response or verified resolution. Missing or invalid durations remain null and are excluded from duration means while affected requests remain counted. Status and closure values reflect the October 2026 capture, not reconstructed status at the historical question date. [2] [11]

Resource-aware execution

Exact row admission precedes CSV-worker reservation. Streaming still enforces byte, row, deadline and identity limits. Normalization bounds output and checks free disk; these observations are not resource reservations. A preview never silently becomes the full export population. [2]

3. Execution and artifact invariants

The service stores specification, catalog and dataset identity with the result. Later artifacts must retain the same selected population even when the conversational preview was truncated. Dataset qualification is checked before comparative execution. [2]

```
discover catalog and immutable dataset identity
validate specification, qualified periods and limits
execute compiled query; save result envelope
load saved result; verify identity and explicit scope
CSV branch:
check exact row admission; reserve background worker
stream stable rows; verify counts; publish CSV + manifest
Maps branch:
verify exact-index data view without implicit time filter
points: use saved DSL; count located / missing rows
trends: publish bounded saved NTA metrics for polygon join
obtain configured Kibana locator
Demo validation: verify actual rendering
```

Different outputs, different evidence

Maps processing does not use the CSV worker. Reusable maps use exact-index data views without implicit time filters. Run 38083900641 verified 242 rendered point IDs. For all 196 represented NTA groups and a selected three-group subset, rendered values matched saved DSL, CSVs and all five joined metrics. Both polygon layers matched all 262 original official geometries. Complete map-validation stages took 13.484, 74.912 and 71.870 s; each browser check respected the unchanged 120 s bound. Eight preceding core failures remain archived. [4] [11]

Interpretation and failure boundaries

All three selected daily-rate changes were positive. The cohort-relative green-to-red palette therefore marks a smaller positive increase in green, not improvement; it is not a zero-centered trend legend. Screenshots and inspected source values provide different evidence. Unsupported requests, unavailable observations and complete zero-count cohorts remain distinct. Missing geography, partial results and failed renders cannot be reported as complete artifacts. [2] [4]

4. Worked example

Capture run 37700939902 retained 1,306,911,416 raw JSONL bytes for 2,133,268 unique requests. Nine core executions reproduced the same 1,582,275,376 normalized bytes and full hash. Captured, normalized, ingested, frozen-index and independent-reference counts all equal 2,133,268. Each execution retains its own index UUID, timings and failures; no records were dropped. [3] [4] [11]

Real-data case	Observed selection	Independent agreement
Brooklyn residential noise, October 1	242 requests	Count, ordered preview and full CSV IDs
All October, borough / agency	336,612 requests; 81 groups	Complete group counts and status totals
October rodent closure, agency	1,935 requests	Count, closed count and valid-duration mean
October, 1 km downtown radius	3,286 requests	Radius selection count and grouping
June / October residential-NTA rodent	3,680 / 1,928 requests; 196 groups	Counts, 30/31-day rates, closure means and ranking

Official geography and analytical scope

The pinned NTA2020 release 26b supplies 262 polygons and 197 residential codes. The comparison contains 196 represented residential areas. Across the full corpus, 32,843 requests lack geometry and 474 coordinates match no supplied polygon. Those records remain visible in quality counts instead of being silently assigned a neighborhood. [4] [12]

The SQL reference shares normalized records and NTA assignments. A separate raw-capture reconstruction additionally verified 9,705 official-polygon assignments and 23 missing locations in the 9,728-row export cohort, using all 262 original polygons. This shares GEOS and timezone libraries, and does not independently qualify every corpus assignment. The ninth run verified all 196 and three selected rendered trend groups; model-led chaining remains a separate outcome. [4] [15]

Table 1. These are observations of the captured public-request corpus. Reporting growth is not incident prevalence, statistical significance or proof that neighborhood conditions worsened.

5. Recorded results and interpretation

Run 38031560273 used Elasticsearch 9.5.5 on a Linux host with four logical CPUs and 16.77 GB RAM. Normalization took 282.763 s; ingestion and freeze took 337.796 s. Query timings include service initialization and saved results, excluding model and browser work. Other retained runs used different host conditions; their timings are not a controlled optimization comparison. [5] [11]

Query family	Median / maximum	Serial repetitions
Filter and preview	28.574 / 31.957 ms	5
Borough / agency aggregation	53.787 / 109.474 ms	5
Administrative closure mean	32.736 / 57.581 ms	5
Geographic radius	33.002 / 41.906 ms	5
Neighborhood period comparison	77.207 / 141.573 ms	5

Five-query batches took 0.201 s at concurrency one and 0.157 s at concurrency two. A full-month Brooklyn residential-noise CSV held 9,728 rows, nine columns and 1,139,937 bytes: 1.060 s individually and 1.112 s for two exports together. Separately, four retained sixth-run exports matched 350,208 exact cells against independently selected raw records, with zero closure error. The files are byte-identical: 87,552 unique logical cells, not four independent samples. [5] [15]

An actual worker exited 86 at first-row streaming. Recovery protected its live lease, then removed abandoned partial output and allowed exact re-export. The recovery call took 0.0181 s; fresh re-export took 0.770 s. This covers one local export-worker failure, not Elasticsearch node or mid-ingestion recovery. [5]

Six samples observed runner RSS up to 127.6 MB, combined worker RSS 58.7 MB and Elasticsearch heap use 1,474.3 MB within a configured 2 GiB heap. These are sampled observations, not guaranteed peaks; node virtual memory is not RSS. Kibana and ingestion peak memory remain unmeasured. [5]

Table 2. This workload lasted 5.034 s after prior index queries, with uncontrolled caches. Five repeats and two short batches establish neither tail latency, sustained throughput, maximum scale nor an SLA.

6. Completed agent evaluation: negative result

All 120 trials executed and received semantic review. One passed automatic and end-to-end checks (0.83%): Q38, repetition 1, correctly refused an unsupported forecast. No other question achieved a pass; the fixed 90% and every-question gates failed. [16]

Only 56 trials delivered finals; 52 timed out, 11 stopped after repeated invalid calls and one produced an invalid final. All nine neighborhood-to-agency chains failed. Reviewers identified unsupported claims or false completion in 32 trials; a separate automatic numeric-evidence check recorded eight hard failures. Two refusal answers passed semantic review, but repetition 2 failed its required automatic status. [16] [17]

Frozen execution and retained failures

Before question exposure, development run 38028703734 failed a 32-record count; the first real-corpus campaign then timed out on all three metadata-only startups at 300 s, exposing no questions. Those failures remain archived. The prospectively amended startup cap was 900 s. The final campaign used Qwen3-1.7B Q4_K_M, native OpenBLAS, 24,576-token context and unchanged 160 s trial, 120 s request and 24-call limits. [8] [13] [14]

Three host jobs each ran the same 40 AI-authored questions in fresh contexts, without reference answers. The merger required equal source, normalized/database hashes, cases, catalog, candidate code and policy; each trial retained its own qualified index, runtime and warmup identity. Reviewers worked from unscored cards, then combined semantic judgments with frozen automatic checks. [14] [16]

Whole-trial median was 126.159 s and maximum 160.006 s across all 120 attempts, including failures and model/tool/adapter work. Startup, download and build are excluded. Separate metadata warmups took 295.782, 256.061 and 289.676 s. These are attempt timings, not generation-only latency or a successful-answer SLA; host differences prevent attributing warmup success solely to the budget change. [17]

7. Validity, qualification and remaining work

Gate	Recorded outcome	Qualification boundary
Acquire and reconcile	2,133,268 observed requests; nine pre-study cores	Three additional study core preflights also passed
Backend and CSV	Five families; 350,208 exact CSV cells; worker recovery	Cell audit covers one cohort; finite workload only
Maps and actual agent	Three Maps cases pass; agent 1/120 (0.83%)	All nine chains fail; conversational release unqualified
Capacity and deployment	Finite Linux workload; Windows r4 installs	No broader-device, sustained-load or billion-row claim

Threats to validity

Matching source counts, revisions and local hashes qualify an observed public-data window; they do not establish provider transaction isolation or complete city reporting. Complaint rates reflect access and reporting behavior. Closure timestamps are administrative records, and their later capture does not reconstruct a historical status snapshot. Missing geometry limits spatial inclusion. [3] [4] [12]

The SQL oracle shares normalization and polygon assignments. The separate raw-to-CSV audit narrows that dependence for one 9,728-row cohort, sharing GIS/timezone libraries; no observed DST ambiguity, invalid or negative closure, escaping or boundary-edge cases occurred. Five analytical families and the 5.034-second workload do not establish sustained load or tail latency. Sampled memory can miss peaks; worker recovery is narrower than node recovery. [4] [5] [15]

Development fixes and the startup amendment preceded question exposure; candidate code, cases and thresholds then stayed frozen. Three seeds on three hosts are repeated observations of 40 questions, not 120 independent questions; host and seed are confounded. The three AI reviewers contributed infrastructure, but did not generate the answering-model completions. Independent human labels, blinded model selection, a second agent and user-time comparisons remain absent. [7] [14] [16]

Conclusion

The service demonstrated reproducible execution on over two million real requests, independently checked export cells, local-worker recovery and three matching rendered Maps cases. The tested local agent failed severely: 1/120 end-to-end passes and no successful analytical question. Reliable conversational use needs a different qualified agent or subsequent development followed by a new evaluation; these results do not qualify the current release. [16] [17]

8. Reproducibility and research record

Nine archived pre-study cores retain separate commits, index identities and failures; Table 2 fixes run 38031560273 as its measured condition. Three additional matrix cores, later regression, r4 installations and agent receipts bind their own sources. Published 488c57c remains the historical edition. Final citations bind the evidence commit recorded in this paper's provenance. [4] [6] [7] [11] [16]

The final Windows suite collected 576 tests: 575 passed, one unprivileged-symlink test skipped, in 66.672 s. Fresh offline core/MCP/geometry r4 installations checked 36 wheels, 3,959 RECORD entries, 23 modules and five resources. Target: CPython 3.12 Windows x64 on an ARM64 host; other-device qualification is unclaimed. [6] [7]

Default-five previews preserve explicit larger requests, exact totals and full CSV membership. The r4 wheels pin packaged core; later repository-only Maps and study tools are separate. Local installation success does not establish million-record laptop performance or agent quality. [7] [8]

```
python -m analytics311 describe
python -m analytics311 mcp
python tools/run_real_acceptance.py --help
python -m tools.agent_campaign --help
```

Transfer ZIPs include core and five historical harness files. New evaluation scripts require the matching repository plus separate data, browser, model and server runtimes. [7]

A reproducible research record

The public nested project source contains implementation, documented contracts and retained development evidence. The publication evidence manifest records SHA-256 fingerprints for each local source cited in this paper. External references link to primary publications or official technical documentation, verified for this edition.

A future rerun should record the exact commit, dependency/runtime identities, input hashes, configuration and environment, plus each attempt and its outcome. Keep generated outputs separate from source and mark any changed implementation as a new condition. Reproduction of a software check does not automatically reproduce a model judgment, benchmark distribution or independent human label.

Authorship and project status

Boomer Rawlings developed this work individually after the original submission and grading dates. The current project is In progress. The implemented methods and recorded observations in this paper are separated from the explicitly proposed research protocol and roadmap. No group-project, graded-submission or completed-research claim is made.

References

Repository references identify the evidence used for the implementation and result claims. Primary literature supports the research framing; its published results are not attributed to this project.

[1] Implemented service and version
IMPLEMENTATION.md

[2] Analytical and artifact contract
docs/tool-contract.md

[3] Actual capture: local streaming verification
examples/evidence/captured-local-verification.json

[4] Ninth real-corpus execution: independent Maps and core review
examples/evidence/real-core-38083900641/independent-review.json

[5] Eighth Elasticsearch workload: actual measurements
examples/evidence/real-core-38031560273/measurements.json

[6] Final v0.7 software regression: 575 passed, one Windows skip
examples/evidence/test-report-v0.7-matrix.json

[7] Verified offline distributions and transfer limits
docs/distribution.md

[8] Actual local-model development quality failure
examples/evidence/model-smoke-38028703734/independent-review.json

[9] Yu et al. (2018). Spider. EMNLP.
https://arxiv.org/abs/1809.08887

[10] Elastic. Paginate search results. Accessed October 9, 2026.
https://www.elastic.co/docs/reference/elasticsearch/rest-apis/paginate-search-results

[11] Acceptance ledger: executed outcomes and qualification limits
docs/acceptance.md

[12] Official NTA2020 release 26b provenance
examples/evidence/official-nta2020-26b.json

[13] Three real-corpus startups with zero question exposure
examples/evidence/agent-startup-failure-38029812736/independent-review.json

[14] Prospective three-replica agent campaign protocol
docs/agent-campaign-v1.json

[15] Independent raw-to-CSV nine-column cell audit
examples/evidence/csv-cell-audit-38029812736.json

[16] Completed 120-trial campaign: strict automatic and semantic merge
examples/evidence/agent-campaign-38084773557/final-review.json

[17] Actual trial failures, whole-trial latency and per-host diagnostics
examples/evidence/agent-campaign-38084773557/actual-diagnostics.json